Graduate Certificate in Artificial Intelligence with Machine Learning
AIGC 5002 - Machine Learning and Deep Learning

Lab 1: Python Basics and Installing Anaconda

Submission guidelines:

- For this lab, you will need to submit 1 PDF file.
- After you complete all the exercises, convert your Jupyter Notebook (.ipynb) to PDF. Name the PDF as follows: firstname_lastname_LAB1.pdf
- Go to the course Blackboard → Labs folder → Lab 1 and submit the pdf.

Part 1: Anaconda Installation and Jupyter Notebook Basics

Please do not hesitate to ask me any questions throughout this lab.

Installation

Before completing the steps below, please visit https://support.microsoft.com/en-ca/help/15056/windows-32-64-bit-faq to determine if your Windows computer is a 32-bit or 64-bit system. This way, you can download and install the correct software. For other operating systems, please research online to help determine if your computer is a 32-bit or 64-bit system.

1. Download **Python 3.9.1** or the latest version from the following link and then install the program using default settings: https://www.python.org/downloads/

2. Download **Anaconda Individual Edition** from the following link: https://www.anaconda.com/

3. Restart your computer after successfully installing Python and Anaconda to update the software settings on the system.

4. Open **Anaconda Navigator** (under the *Anaconda3* folder in the Start menu on Windows computers) and launch **JupyterLab**. You may be asked to choose an application to open JupyterLab. If this happens, select a Web browser such as Google Chrome, Mozilla Firefox, or Safari. **(Chrome is strongly recommended)**

5. In the *JupyterLab* launcher window, browse to the downloaded (from Blackboard → Labs folder → Lab 1 "*download the files*") Jupyter Notebooks (.ipynb) file on your computer's hard drive using the navigation options available on the left-hand side of the window.

6. You **must not** submit these files. Instead, use them as a reference to Jupyter basics and Python revision.

7. Visit the link at https://jupyterlab.readthedocs.io/en/latest/user/interface.html# to familiarize yourself with the JupyterLab interface.

Saving Files in JupyterLab

1. When downloading a dataset to your computer, make sure it is saved in the same directory as your Jupyter Notebook (IPYNB) file to avoid receiving errors when retrieving its contents. (Otherwise, make sure you access the files using the absolute file path.)

2. To save your Jupyter Notebook file, open the **File** menu and select **Save Notebook**.

3. To save your Jupyter Notebook (.ipynb) file as a PDF, follow the steps below:

   a. Open the **File** menu, select **Print...**, choose **Save as PDF** from the list of available printers on your computer select a large paper size (A2, A3), and then click **Save**.

   b. In the *Save as* window, rename the PDF file. Please select where to save it on your computer, then click **Save**.

   c. Browse to where you saved the PDF file and open it to ensure it was created successfully.

4. When finished using JupyterLab, open the **File** menu, select **Close and Shutdown Notebook**, click **OK** to confirm the shutdown of the Notebook, re-open the **File** menu, select **Log Out**, and then close the *Jupyter Notebook* browser tab.

5. In the *Anaconda Navigator* window, open the **File** menu, select **Quit**, click **Quit** again to close any running applications, and click **Yes** to quit Anaconda Navigator.

Part 2: Python basics

Revision:

- Create a new Jupyter Notebook. You will convert this file to PDF at the end and then submit it.

- Control Structures:
  - Create a loop that prints numbers from 1 to 10.
  - Create a loop that prints even numbers between 1 to 10.
  - Use an if-else structure to check if a number is odd or even.

- Functions:
  - Write a function that takes a number as an argument and returns its square.
  - Write a function that takes a string as an argument and returns its reverse.

Exercises:

1. Write a Python function named second_largest that takes a list of integers and returns the second-largest number in the list. Assume the list has at least two distinct elements.

   Sample Input: [5, 2, 8, 1, 7]
   Sample Output: 7

2. Write a function named accumulate_squares that takes an integer n and returns the sum of the squares of all numbers from 1 to n that are not divisible by 3.

   Sample Input: 5
   Sample Output: 46

Enjoy!